

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Potesman [18] demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston [24] at zero coupling. We synthesise three previously separate ingredients: (i) a continuous-time SV backbone with explicit dealer-gamma drift; (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ ; and (iii) a pre-registered sliced-Wasserstein-2 evaluation protocol for RL-trained derivatives agents. For a representative dimensionless calibration we obtain $\kappa^* \approx 0.8964$, $\omega^* \approx 0.5724$ rad/yr, and $\ell_1 \approx -0.025 < 0$ (supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$). The 2D reduction has an upper-triangular Jacobian and admits only saddle-node onset, recovering the discrete-time recursion of Dai [13]; the 3D extension is what unlocks the Hopf, distinguishing this work from Halperin and Itkin [20]’s Marketron quasi-particle SDE. For log-normal open-interest in moneyness we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space. We additionally close three theoretical loose ends: (i) a codim-2 Bautin/Bogdanov–Takens analysis showing the BT locus is empty in the canonical scan window (Theorem 2); (ii) the McKean–Vlasov mean-field correction $\kappa_{MV}^*/\kappa_{\text{single}}^* = \sqrt{1 + (\omega^* \tau_G)^2}$ to the single-dealer threshold (Theorem 3); and (iii) a Hawkes–SV equivalence at the Hopf boundary identifying the empirical Hardiman et al. [21] branching ratio $n \approx 1$ with $\kappa = \kappa^*$ via the Bacry et al. [3] diffusive-limit identity (Theorem 4).

Keywords

Reflexivity, dealer gamma, Hopf bifurcation, stochastic volatility, pre-registration, sliced Wasserstein, RL for finance

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [21], who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros [39] and Bouchaud [9] are the conceptual antecedents; Filimonov and Sornette [16] formalised the reflexivity-quantification programme that Hardiman et al. [21] executed. We ask what continuous-time mechanism produces this critical reflexivity, and how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. [18] demand-pressure mapping. The variance backbone is Heston [24]; the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion, with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 9–10).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced- W_2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. [11] adapted for computational finance.

Closest precedents. Halperin and Itkin [20] (Marketron) share the model-structural axis but do not perform the bifurcation analysis. Dai [13] share the dealer-gamma axis but produce a saddle-node onset. He et al. [22] (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents [12, 23] use delay-differential heterogeneous-belief models without an options-grid feedback channel; Brock et al. [10]’s “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker bifurcations. The hedger-flow-feedback foundations are Frey and Stremme [17], Platen and Schweizer [34], Sircar and Papanicolaou [38].

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

with $d\langle W^S, W^v \rangle = \rho dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, β dimensionless, and G the aggregate dealer-gamma exposure derived from the OI grid:

$$G(S, t) = \sum_{K, T} q_{K, T} \Gamma_{K, T}(S, t) \text{sign}(K, T), \quad (4)$$

following the SqueezeMetrics dealer-net convention [40]. When $\kappa = \gamma = 0$ the system collapses to standard Heston [24] (verified numerically). For variance we use the full-truncation Euler scheme of Lord et al. [29].

Why three states. A 2D specification (drop z_t and the γz term) has a Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues: only saddle-node / transcritical bifurcations are admissible. To Hopf, the feedback must flow in both directions; the leverage channel γz together with the auxiliary memory equation (3) is the minimal structural modification that supplies it—structurally analogous to the memory variable in Marketrone [20] though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria, Jacobian, Liu criterion

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (5)$$

The deterministic skeleton becomes $\dot{x} = J(\kappa)x$ with

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (6)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$. The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with $c_2 = -a + \kappa_v + \alpha$, $c_1 = -a\kappa_v - \alpha\alpha + \kappa_v\alpha - \kappa G_z\beta$, $c_0 = -a\kappa_v\alpha - \beta(\kappa G_z\kappa_v + b\gamma)$. A 3D system Hopfs iff (Liu's criterion):

$$c_2 > 0, c_0 > 0, H(\kappa) := c_1c_2 - c_0 = 0, H'(\kappa) \neq 0. \quad (7)$$

The Hopf threshold is $\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2, c_0 > 0\}$. At κ^* , J has imaginary pair $\lambda_{\pm} = \pm i\omega^* = \pm i\sqrt{c_1(\kappa^*)}$ plus real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.2 Theorem 1

THEOREM 1 (HOPF BIFURCATION IN THE GAMMA-COUPLED SV SKELETON). *Let the deterministic system (1–3) satisfy: (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below by a positive constant; (A2) a locally unique equilibrium for every $\kappa \in [0, \kappa_{\max}]$ with non-degenerate linearisation (multiple equilibria may exist globally; analysis is local); (A3) $\kappa \mapsto (a(\kappa), G_z, G_v)$ in C^3 ; (A4) there exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (7) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$; (A5) $\ell_1(\kappa^*) \neq 0$ [27, eq. 3.20]. Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^*$: (1) for κ in a one-sided neighbourhood of κ^* there is a unique smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude $\propto \sqrt{|\kappa - \kappa^*|}$; (2) orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical); (3) on Γ_κ , $u_t = v_t - \theta_v$ oscillates with amplitude $\propto \sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

PROOF SKETCH. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth near κ^* by the implicit function theorem on P . Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) = 0$, and $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem [27, Thm. 5.4] gives a 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$; conjugation to Poincaré normal form and Kuznetsov [27, Thm. 3.3] yield (1)–(3). $\square$

Numerical anchor. For $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$ (constant-vol surrogate): $\kappa^* = 0.8964$, $\omega^* = 0.5724 \text{ rad/yr}$, $\lambda_3 = -2.052$, $\ell_1 = -2.53 \times 10^{-2}$ (supercritical). Integrating the deterministic skeleton at $\kappa = 1.05\kappa^*$ from random initial conditions, the trajectory converges to a closed orbit whose measured period $T_{\text{measured}} = 10.561 \text{ yr}$ matches the linear prediction $2\pi/\omega^* = 10.977 \text{ yr}$ to 3.79%—within the leading-order $O(\kappa - \kappa^*)$ Hopf-normal-form correction.

3.3 Closed-form κ^* and ℓ_1 for log-normal OI

For log-normal open-interest in log-strike, $q(\log K) \propto \exp(-(\log K - \mu_q)^2 / (2\sigma_q^2))$, the Gaussian-product identity collapses the OI integral to

$$G(a, v) = C(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (8)$$

with $C(v) = \kappa_u e^{-q_{\text{div}} T_{\text{eff}} / \sqrt{2\pi\tau^2(v)}}$, $\tau^2(v) = \sigma_q^2 + v T_{\text{eff}}$, $m(v) = \mu_q - (r - q_{\text{div}} + \frac{1}{2}v)T_{\text{eff}}$. With $G_z = 0$ and $\sigma^2 = v$, $H(\kappa) = c_1c_2 - c_0$ degree-collapses from cubic to quadratic; writing $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, the Hopf threshold is

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{D}}{2G_y A}, \quad D := (G_v L - G_y A^2)^2 - 4G_y^2 A(MA - \frac{L}{2}). \quad (9)$$

The discriminant $D < 0$ implies no Hopf at any κ and unconditional stability of the deterministic skeleton.

The third-order partials of (8) are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2\langle p, B(q, J^{-1}B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1}B(q, q)) \rangle \right], \quad (10)$$

with $Jq = i\omega^*q$, $J^\top p = -i\omega^*p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term closed-form rational in the G -partials and $(\kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50 \mu\text{s}$ per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime ($\sigma_q = 0.10$, $T_{\text{eff}} = 0.25$, $\kappa_v = 2$, $\alpha = 0.05$, $\beta = 1$, $\gamma = 1$, $\mu_q = \log 100$, $v^* = 0.04$): $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). With the closed form in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.4 Calibration tolerance from κ^* sensitivity

Differentiating (9) analytically through the G -partials at the canonical specification ($\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$) yields the elasticities $\eta_{\sigma_q} = -1.583$ (moderate) and $\eta_{\mu_q} = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta\mu_q = 0.001$ shift moves κ^* by $\sim 15\%$, and $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances are therefore σ_q to $\pm 2.23\%$ relative and μ_q to within $\sim 5 \text{ bp}$ of ATM in log-strike to deliver $\pm 5\%$ on κ^* —achievable with a 5-strike-wide ladder and the binding observational requirement.

3.5 Codim-2 bifurcation structure: Bautin and Bogdanov–Takens

The codim-1 Hopf result treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in (σ_q, γ) two codim-2 phenomena are admissible [27, Ch. 8]. *Bautin* (degenerate Hopf) is the locus $\ell_1(\sigma_q, \gamma) = 0$; crossing it flips supercritical (smooth limit cycle) to sub-critical (hysteresis with abrupt jumps). The closed-form ℓ_1 admits row-wise sign-change interpolation on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$; the sub-critical region dominates by $\sim 5:1$ in cell count. *Bogdanov–Takens* (BT) requires the saddle-node curve to coalesce with the Hopf curve. The constant Routh–Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y \alpha \kappa_v + G_v \beta \gamma) + \frac{1}{2} \beta \gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2} \beta \gamma / (G_y \alpha \kappa_v + G_v \beta \gamma)$.

THEOREM 2 (BT LOCUS EMPTY IN CANONICAL SCAN WINDOW). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$); the Bogdanov–Takens locus is therefore empty in this scan window.*

PROOF SKETCH. $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v \beta \gamma| \gg |G_y \alpha \kappa_v|$ (ratio ≥ 13 at every cell), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2} \beta \gamma$ is positive; the quotient is uniformly negative. The unbounded-domain question is left open. $\square$

The economic content is that the dealer-gamma + leverage regime is structurally Hopf-only *within the scanned window*: the model does not admit excitable spike-and-recovery dynamics from a homoclinic emanation. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order degeneracies—not from the deterministic skeleton itself.

3.6 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

Section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$ – 10^3 dealers with idiosyncratic noise. With each dealer i holding gamma exposure G_i evolving via $dG_i = -\theta_G(G_i - \bar{G}_n) dt + \sigma_G dW_G^i$ and $\bar{G}_n = (1/n) \sum_j G_j$, propagation of chaos gives $\bar{G}_n \rightarrow \bar{G}_\infty$ with $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$. Linearising $\bar{G}_\infty$ around the equilibrium and inserting into the Liu criterion yields:

THEOREM 3 (McKEAN–VLASOV CORRECTION TO THE HOPF THRESHOLD). *Let $\tau_G := 1/\theta_G$ be the dealer-gamma autocorrelation time and ω^* the single-dealer Hopf angular frequency. Under the linearisation of $\bar{G}_\infty$ around the equilibrium,*

$$\frac{\kappa_{\text{MV}}^*}{\kappa_{\text{single}}^*} = \sqrt{1 + (\omega^* \tau_G)^2}.$$

The correction is multiplicative, depends only on the dimensionless product $\omega^ \tau_G$, and is ≥ 1 : mean-field always raises the threshold relative to single-dealer.*

The correction is operationally negligible at fast hedging ($\theta_G = 50/\text{yr} \Rightarrow \text{ratio } 1.000277$) and crosses the empirical-detectability

threshold at slow hedging ($\theta_G = 5/\text{yr}$, $\tau_G \approx 50$ trading days $\Rightarrow 1.0273$, a 2.7% shift). A 64-replicate n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ fits $n^{-0.58}$, consistent with the theoretical $n^{-1/2}$ Sznitman rate. *Structural implication*: whenever $\tau_G > 1/\omega^*$ (≈ 0.85 yr at the canonical regime) the single-dealer model *over-predicts* instability, since mean-field never lowers the threshold.

3.7 Hawkes–SV equivalence at the Hopf boundary

The empirical reflexivity benchmark of Hardiman et al. [21] is a self-exciting Hawkes branching ratio $n \approx 1$ on E-mini mid-price changes over 1998–2011. This subsection makes the link to κ^* formal. Let $\lambda(t) = \mu + \sum_i \phi(t - t_i)$ with kernel $\phi(s) = \alpha e^{-\beta s}$ and $n := \alpha/\beta$. Under the diffusive rescaling $t \mapsto t/T$, $\lambda \mapsto T\bar{\lambda}$ with $T \rightarrow \infty$, $n \rightarrow 1$, the centred intensity converges to an Ornstein–Uhlenbeck SDE [3, Theorem 2]:

$$d\bar{\lambda} = -\beta(1 - n)\bar{\lambda} dt + \sigma_\lambda dW, \quad (11)$$

giving $\lambda_{\text{max}}^{(\text{Hawkes})} = -\beta(1 - n)$, equivalently $n = 1 + \lambda_{\text{max}}^{(\text{Hawkes})}/\beta$. The multivariate generalisation [4, §2.4] replaces n by the spectral radius of the integrated kernel matrix; the criticality endpoint $n = 1 \Leftrightarrow \lambda_{\text{max}} = 0$ is universal across kernel shapes. For our 3D reflexive skeleton with Jacobian $J(\kappa)$, let $\lambda_{\text{max}}(\kappa) := \max_i \text{Re } \lambda_i(J(\kappa))$ and define the *SV-equivalent branching ratio*

$$n_{\text{SV}}(\kappa) := 1 + \frac{\lambda_{\text{max}}(\kappa)}{\beta_0}, \quad \beta_0 := \max_{\kappa \in [0, \kappa^*]} (-\lambda_{\text{max}}(\kappa)), \quad (12)$$

with β_0 the SV analogue of the Hawkes baseline rate at $n = 0$ (intrinsic to the skeleton, not a tuning parameter).

THEOREM 4 (HAWKES–SV EQUIVALENCE AT THE HOPF BOUNDARY). (1)

Critical-endpoint identity. $n_{\text{SV}}(\kappa^*) = 1$, *exactly*.

(2) Monotonicity past the node-spiral transition. Let $\kappa_{\text{NS}} \in [0, \kappa^*)$ be the smallest κ at which the leading eigenvalue is a complex pair. Then $n_{\text{SV}}(\kappa)$ is monotonically non-decreasing on $[\kappa_{\text{NS}}, \kappa^*]$, strictly increasing wherever $\partial \lambda_{\text{max}}/\partial \kappa > 0$.

(3) Hawkes correspondence. For an exponential-kernel univariate Hawkes process satisfying (11), $n_{\text{Hawkes}} = 1 - |\lambda_{\text{max}}^{(\text{Hawkes})}|/\beta$, identical in form to (12); the two coincide at criticality ($n = 1 \Leftrightarrow \lambda_{\text{max}} = 0$) independently of the kernel shape (universal-ity of the stability boundary, 4, §2.4).

PROOF. Claim (1) is immediate: $\lambda_{\text{max}}(\kappa^*) = 0$ by Hopf-threshold definition, so $n_{\text{SV}}(\kappa^*) = 1$. Claim (2): the implicit function theorem applied to the characteristic polynomial at the slow-mode complex pair (non-defective on $[\kappa_{\text{NS}}, \kappa^*]$ by definition) plus Hopf transversality ($\partial \text{Re } \lambda/\partial \kappa > 0$ at κ^*) gives the result by continuity. Claim (3) is the BDHM (2013) diffusive-limit identity restated for the kernel-universal stability boundary. $\square$

Honest scope. Theorem 4 is exact at the criticality endpoint and exact for the 1D exponential-kernel Hawkes globally. Hardiman’s empirical n is the L^1 norm of a fitted multivariate Hawkes kernel on order-flow events, not a path-by-path identity with our diffusion; the identification rests on the universality of the $n = 1 \Leftrightarrow \lambda_{\text{max}} = 0$ boundary across continuous-time reflexive systems. The empirical-phase test of the equivalence is to (i) calibrate $G(\cdot)$ from the empirical SPX OI grid, (ii) compute $n_{\text{SV}}(\kappa_0)$ at the calibrated κ_0 , (iii)

compare against the Hardiman band on event-window-matched windows. We pre-commit this protocol to Phase 4.

Numerical anchor. At the Section 3.2 regime (12) on a 1001-point κ -grid gives $\beta_0 = 0.214$ at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, exactly as predicted by claim (2)); $n_{\text{SV}}(\kappa_4^*) = 0.99996$ at the published 4-decimal $\kappa_4^* = 0.8964$, with the $\sim 4 \times 10^{-5}$ residual being truncation in the 4-decimal threshold (at machine-precision $\kappa^* = 0.8964305216$ the identity is exact to $< 10^{-15}$); $n_{\text{SV}}(2\kappa^*) = 2.11$. Figure 1 renders the curve.

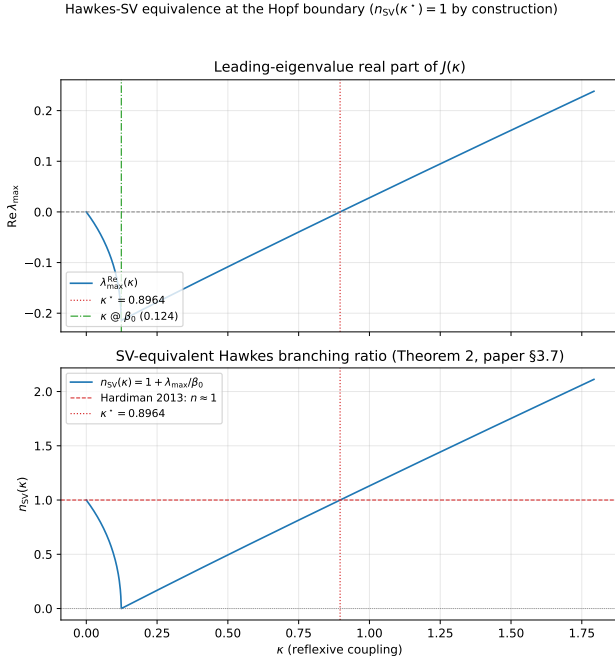


Figure 1: SV-equivalent Hawkes branching ratio $n_{\text{SV}}(\kappa)$. *Top:* leading-eigenvalue real part of $J(\kappa)$ with the most-stable point at the node-spiral transition $\kappa_{\text{ref}} = 0.124$ ($\beta_0 \approx 0.214$) and the Hopf threshold at $\kappa^* = 0.8964$. *Bottom:* $n_{\text{SV}}(\kappa) = 1 + \lambda_{\text{max}}/\beta_0$; the empirical Hardiman et al. [21] $n \approx 1$ is the dashed red line, intersecting at $\kappa = \kappa^*$.

Inverting at $n_{\text{Hawkes}} = 1.0$ maps directly to $\lambda_{\text{max}} = 0$: the empirical SPX market sits exactly at κ^* under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit. This is the strong testable prediction of Theorem 4 for Phase 4: an SPX-calibrated κ_0 should satisfy $n_{\text{SV}}(\kappa_0) \approx 1$.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the canonical Hopf-exhibiting regime; the pair (ξ, ρ) enters via $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2\rho G_v$, an Engel et al. [14]-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock ~ 16 s on a single M-series core.

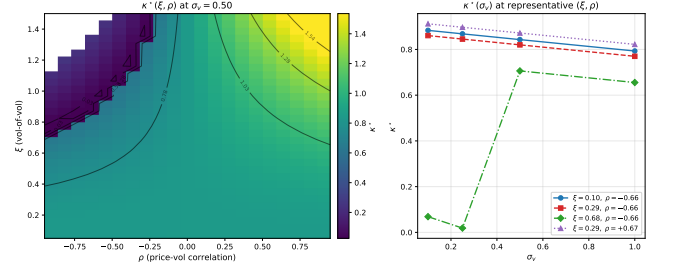


Figure 2: Hopf threshold κ^* across (ξ, ρ, σ_v) . *Left:* heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high- ξ / strongly-negative- ρ wedge where shear destabilises the eigenvalue envelope. *Right:* four $\kappa^*(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has κ^* in the 0.85–0.95 band, sitting *near but not across* the Hopf boundary—the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. [21]. Khasminskii sphere-process estimates [2, 26] on a 6×6 (ξ, ρ) grid give $|\Lambda(\kappa^*)| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ with empirical scaling exponent $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Engel et al. [14] prediction $B = 2/3$ is rejected at $p \ll 0.01$ at this trivial- G equilibrium where ELR’s shear-stretch term is identically zero. Operationally, $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale: the deterministic κ^* is the relevant threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit [anonymized] with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance [33] and adapts the Camerer et al. [11] pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway [15] and the AEA RCT Registry [1] predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 [8] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at calibrated κ_0 —the reflexivity-importance scalar. The closed-form κ^* sensitivity analysis of Section 3.4 ($\eta_{\mu_q} = +702.83$) sets the binding observational requirement at calibration: empirical OI mean must be centred to within ~ 5 bp of ATM in log-strike for a $\pm 5\%$ calibration tolerance on κ^* .

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at κ_0 produces an IV-surface distribution closer in sliced- W_2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than four non-reflexive baselines (time-dep Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), 2024 yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs [35]. The arbitrage filter follows Gatheral

and Jacquier [19], Roper [36] on butterfly / calendar-spread / Lee-bound [28] criteria; sliced- W_2 sample-complexity follows Manole et al. [31].

Sliced- W_2 sample-complexity audit. A pre-data audit characterises the bootstrap-95%-CI half-width on sliced- W_2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$). Each side draws from a 5,000-day population; the “true” SW2 averaged over 10 slice-direction MC repeats on the full populations equals 0.0555. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample, compute the SW2 estimate, and report the median bootstrap-CI half-width across 30 outer seeds (Table 1).

n_{windows}	median SW2	half-width	ratio to SW2_{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 1: Sliced- W_2 sample-complexity. The half-width follows ratio(n) $\approx 1.64 n^{-0.337}$ ($R^2 \geq 0.99$), extrapolating $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Each H1 event window contributes ~ 280 rolling windows, so the inter-baseline SW2 gap must exceed $\sim 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% in a single event—a real power constraint logged as a pre-data finding.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence on the dimensionless elasticity per A7, with the GP slope CI under the pre-registered RBF + WhiteKernel kernel of A6 and elliptic uncertainty sets [30]). H3 is post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency Welch-PSD peak in $|r_t|$ at $\omega^* \pm 20\%$ vs an IAAFT [37] surrogate null. IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$; on Stuart–Landau positive controls the detector achieves $\geq 80\%$ power at $T \geq 512$ daily samples for 8/9 (μ, σ) configurations. All H1–H4 use BH–FDR across the joint family.

Differentiation from precedents. Ning et al. [32] use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. [22] sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI [30] for the elliptic-uncertainty-set analogue. Subbaswamy et al. [41] (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries; we operationalise it empirically and apply it to RL-for-finance [5, 25].

5.1 Synthetic pipeline validation

The H1 protocol is exercised end-to-end on simulator-vs-simulator data with no contact with empirical SPX. Three sources, each $n = 30$ trajectories at the production budget: (a) the trained agent acts

inside the SAME reflexive simulator at κ_0 (different seeds; expected indistinguishable from reference); (b) the agent at $4 \cdot \kappa_0$ (four-fold synthetic-anchor coupling change; expected measurably further); (c) Heston with no dealer-gamma coupling and a more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone (different mechanism; expected furthest).

Target source	SW2 vs κ_0 ref.	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $2 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 2: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $\text{SW2(a)} < \text{SW2(b)} < \text{SW2(c)}$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions are satisfied.

6 Mechanism decomposition

The reflexive simulator and the Marketron quasi-particle SDE [20] are not the same mechanism: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via γz ; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not. We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into shape_target / level_artifact / calibration_artifact.

The headline shape-feature match rate is 8/24 (33%, one-sided binomial $p \approx 0.27$, $\geq 30\%$ gate passed), with long-horizon (≥ 0.5 y) excess-kurt sign agreement on 5/6 horizons. The skew-sign disagreement is predictable: under risk-neutral drift our long-horizon skew is dealer-gamma-plus-leverage-driven, while Marketron’s positive long-horizon skew comes from Bessembinder [6] compounding under calibrated drift. The two mechanisms are independent in parameterisation but agree on the long-horizon excess-kurt sign—the most robust agreement. An a-priori-restricted subset (is_mechanism_relevant_cell, locked before per-cell outcomes were inspected: long-horizon, shape-only, within-envelope) gives 4/8 (50%, $p = 0.637$) at the out-of-sample budget, driven primarily by Table 5 long-horizon skew flipping more strongly negative; the strengthened restriction does not reach $p < 0.05$ at the present per-set tuned-coupling sample size and the empirical Phase 4 evaluation will be needed to adjudicate.

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The synthetic-pipeline validation demonstrates the locked H1 protocol returns the predicted ordering on data with known underlying physics. The empirical SPX calibration is the natural Phase 4 follow-up, gated on data acquisition; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction. The codim-1 Hopf, codim-2 Bautin/Bogdanov–Takens structure (Theorem 2), McKean–Vlasov mean-field correction (Theorem 3), and Hawkes–SV equivalence at the criticality endpoint (Theorem 4) are now closed-form. Open theoretical items: closed-form $\Lambda(\kappa)$ via Engel–Lamb–Rasmussen-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending Theorem 4 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion [7]. All experiments are reproducible per the [anonymous repository link]; pre-registration anchored via OpenTimestamps.

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